IoT based Traffic Light Control System using Raspberry Pi

Elizabeth Basil
Department of Electronics and Telecommunication
NBNSSCOE, Pune, India.
liz7beth@gmail.com

Prof.S.D.Sawant
Department of Electronics and Telecommunication
NBNSSCOE, Pune, India.
sd_sawant123@gmail.com

***Abstract*— Congestion in traffic is a serious issue. In existing system signal timings are fixed and they are independent of traffic density. Large red light delays leads to traffic congestion. In this paper, IoT based traffic control system is implemented in which signal timings are updated based on the vehicle counting. This system consists of WI-FI transceiver module it transmits the vehicle count of the current system to the next traffic signal. Based on traffic density of previous signal it controls the signals of the next signal. The system is based on raspberry-pi and Arduino. Image processing of traffic video is done in MATLAB with simulink support. The whole vehicle counting is performed by raspberry pi.**

Keywords—Vehicle Counting;Traffic Light Control;Image Processing;Raspberry Pi;IoT.

I. INTRODUCTION

Internet of Things (IOT) is an important research area which allows a strong connection to communicate between the physical objects through internet connectivity. The Internet of things is the interconnection between the physical objects such as cars, buildings, other vehicles, kitchen appliances and other embedded devices with electronics, software, sensors and actuators that enables these objects to send and receive data over a network without the requirement of human-to- human or human-to-computer interaction. Internet of things device is a device stand-alone internet connected device that can be remotely controlled from other location.IOT based systems have been proposed in several applications such as managing diabetes therapy , building smart home ,supporting disabilities, improving safety in mining operations. In order to use IOT for intelligent relationship applications it is used with a Ubiquitous Computing [1].

Traffic congestion is increasing day by day and we have to face with many problems. Main reasons for traffic congestion are high volume of vehicles, inadequate infrastructure, and the irrational distribution of the development a. Traffic congestion is characterized by longer trip times, slower speeds, , and increased vehicular queuing. When the traffic demand is high that the interaction between vehicles slows the speed of the traffic stream, these also results in some traffic congestion [2].The conventional systems are based on timer which causes limitations such as wastage of time, heavy traffic jam, even though there are no vehicles on the road, still signal is red because of static timings of timers of the signal [3].

Architecture of the Control System

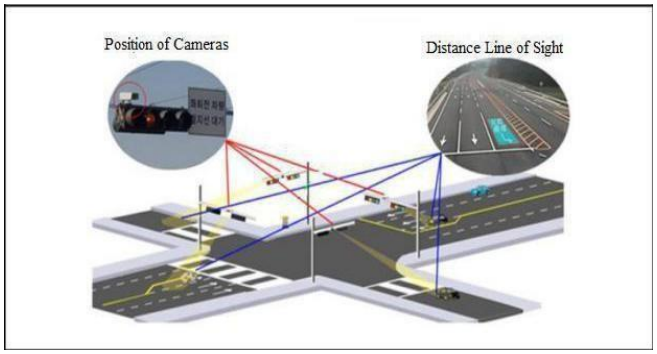


Fig.1. System Architecture

In this architecture camera is placed on the top of the signal so as to capture the traffic video on the particular side of the signal so that it can analyze the traffic in that particular side and get the vehicle count. Based on this count the density of that particular side will be determined and corresponding signal will be provided as shown in figure 1.

In the proposed system following are the design objectives

- Distinguish the presence and absence of vehicles in road images.
- Signal the traffic light to go red if the road is empty
- Signal the traffic light to go red if the maximum time for the green light has elapsed even if there are still vehicles Present on the road.

This paper is organized as follows: Section II gives the motivation about IoT based traffic control system. Information about the methodology used. How the system works is described in Section III. Software design, Hardware components and its implementation is discussed in Section IV. Section V describes results of proposed system. Conclusion of the proposed work is given in Section VI.

978-1-5386-1887-5/17/$31.00 ©2017 IEEE

Authorized licensed use limited to: VIT University. Downloaded on August 15,2025 at 07:08:39 UTC from IEEE Xplore. Restrictions apply.

II. MOTIVATION

Almost all urban cities in the world use traffic light signals to control the traffic on the roads. Different types of traffic light control systems are developed which are vehicle actuated lights and static traffic lights. But their traffic lights timing are fixed and switching patterns are also predefined in the system it is independent of traffic conditions for the different lanes and they are not changing with real time data. They do not consider the non-uniform changing patterns of traffic density throughout the day. It does not matter during a particular period of time route one has more cars than route two; the green light allocation time and pattern still remains the same for all routes. The lack of intelligent techniques in these devices does very little in improving the road network performance and traffic congestion levels. In proposed system unnecessary waiting time in the traffic signal can be avoided by determining traffic density and deciding in which the green signal should be large during the traffic.

III. PROPOSED WORK

A. Block Diagram of Proposed Scheme

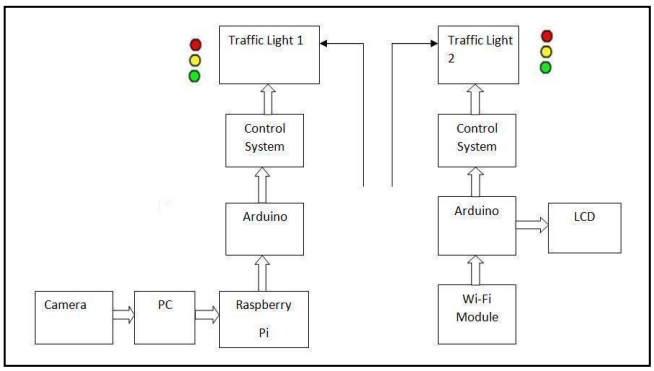


Fig.2. Block Diagram of the proposed system

In existing systems time is already assigned to each signal. According to assigned "time" signals work in each lane. Suppose there are two lanes L1 and L2. Suppose, All vehicles in lane (L1) are passed and vehicles in another lane (L2) are still in waiting state because signal is still red. These systems are static and they cannot handle the various simple traffic conditions throughout the day. In this type of system there is unnecessary waiting time. We propose a traffic signal control system in which based on vehicle density signal time varies. It consists of ESP8266 serial WI-FI wireless transceiver module which sends vehicle counting data to the next signal based on this data its signal timings are updated. The system is based on raspberry pi and arduino. Camera captures the video. It is send to PC and raspberry pi for image processing and vehicle counting. It also consists of arduino microcontroller which controls the signal timings to which LED's are connected. System is implemented in MATLAB & Simulink.

- Steps in the proposed system for controlling traffic light 1:

1. Camera: Continuously record traffic video.
2. Read Image: Read frames of the traffic image.
3. Grayscale Image Conversion: It converts color image to grayscale image. This method is based on different color transform. According to the R, G, B value in the image, it calculates the value of grayscales and convert the image into grayscale image.
4. Image Binarization: Grayscale image is converted into black and white image.
5. Blob Analysis: Blob analysis is performed on the binary image in order to get vehicle count in SIMULINK. In blob analysis we find area of the object. Remove Small objects in the image. Convert that image into logical image in simulink. Perform labelling of object to get counting of the vehicles.
6. Traffic Signal Control: Based on vehicle count signal timings are changed and the respective LED glows.

- Steps for controlling traffic light 2:
  1. Initialize System
  2. Configure ESP 8266 module for multi access point through AT commands
  3. Connect WI-FI module to WI-FI network
  4. Start UDP local port in WI-FI module
  5. Establish UDP connection to Raspberry pi
  6. Wait for data
  7. Change traffic light signal 2 depending upon their received data from raspberry pi

B. Software Design

1. Arduino IDE

The Arduino Software (IDE) is a open source compiler in which we can write code and upload it to the board. It runs on Mac OS X, Linux and Windows. It is based on other open-source software and its environment is written in Java. Arduino compiler accepts C and C++ many of the libraries are written in C++.The Arduino environment performs some small transformations on the written code to make sure that the code is correct then it gets passed to a compiler AVR-GCC, which turns the code into the language which is compatible with the arduino.

2. MATLAB with Simulink support

MATLAB is a high-performance language its basic data element is an array that does not require dimensioning. Matrix and vector formulations allows us to solve many technical computing problems. It combines programming visualization with computation which provides easy-to-use environment where we can express our problems and solutions in the mathematical form. MATLAB toolboxes are available in various fields including control systems,

Authorized licensed use limited to: VIT University. Downloaded on August 15,2025 at 07:08:39 UTC from IEEE Xplore. Restrictions apply.

simulation, signal processing, neural networks, wavelets, fuzzy logic, and many others. Simulink is developed by MathWorks it is a graphical programming environment for simulating, modeling and analyzing multiple domain dynamic systems.

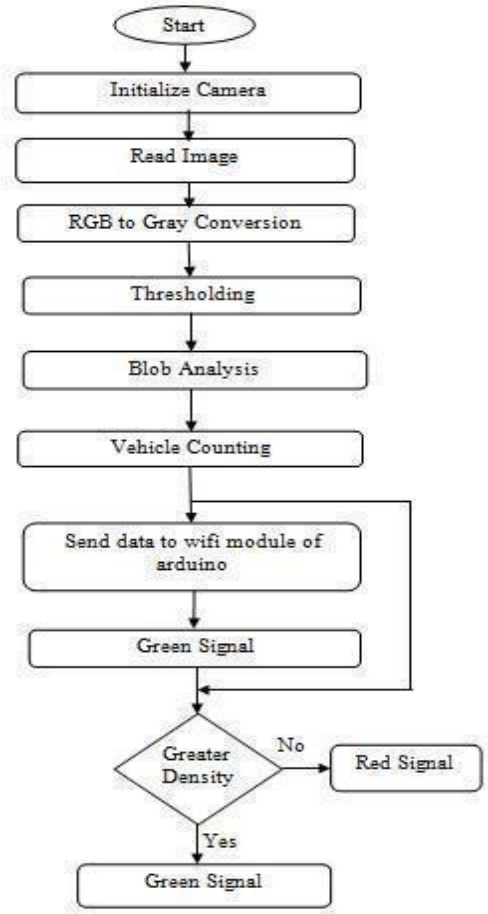


Fig.3. Proposed System Flow

IV. Hardware Implementation

1. Raspberry Pi

Raspberry pi is based on Broadcom system on a chip (SoC), which consists of on-chip ARM compatible central processing unit (CPU) and an on chip GPU(Graphics Processing Unit). Raspberry pi has CPU speed which ranges from 700 MHz to1.2 GHz. Most boards have HDMI composite video output, a3.5 mm audio phone jack and one and four USB slots. It has on board memory range from 256 MB to 1 GBRAM. In raspberry pi operating system is stored in SD cards. Through number of GPIO pins Lower level output is provided that support common protocols like I²C. Pi 3 and Pi Zero W have on board Bluetooth and WI-FI 802.11n.

2. Arduino Uno

The Arduino Uno microcontroller board is based on ATmega328P which has 14 digital input/output pins in which 6 are analog input pins, 6 can be used as PWM outputs), a USB connection, a power jack ,a 16 MHz quartz crystal, ,a reset button and an ICSP header. It allows the transfer of data through serial communication to PC with a USB cable. The board can be powered through a AC-to-DC adaptor battery to get started.

3. Wi-Fi Module - ESP8266

The ESP8266 Wi-Fi Module is a SOC with integrated TCP/IP protocol stack which allows any microcontroller to access Wi-Fi network. ESP8266 module is pre-programmed with an AT command set firmware. It can host an application or offload all Wi-Fi networking functions from another application processor. This module has storage capability and a powerful on-board processing that allows integration of this module with the sensors and other application specific devices through its general purpose I/O pins.

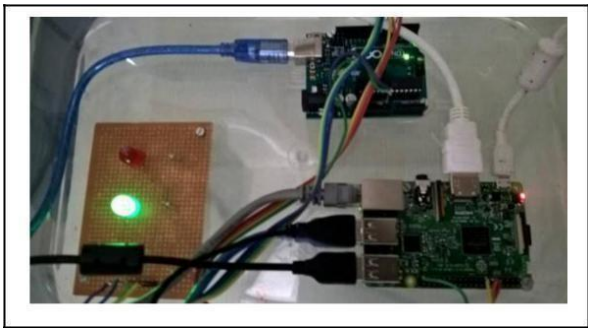


Fig.4. Implemented Model

V. Implementation Results

A. Object Detection

Vehicles detected in the input image are shown in figure 5. It is the binary image with white objects and the black as its background.



Fig.5. Binary Image with vehicles detection

B. Vehicle Counting Results

After applying blob detection algorithm on binary image we get vehicle count on the road. Below figure 7 shows the vehicle count.

Authorized licensed use limited to: VIT University. Downloaded on August 15,2025 at 07:08:39 UTC from IEEE Xplore. Restrictions apply.



Fig.6. Result of object removal

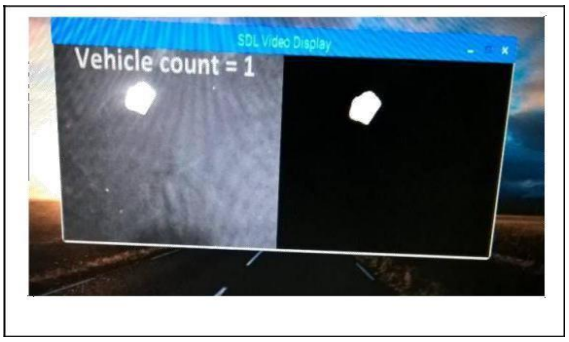


Fig.7. Results of vehicle count

C. Results of data send through WI-FI

Below figure 8 shows status of received data which is transferred through UDP to the WI-FI module of the arduino

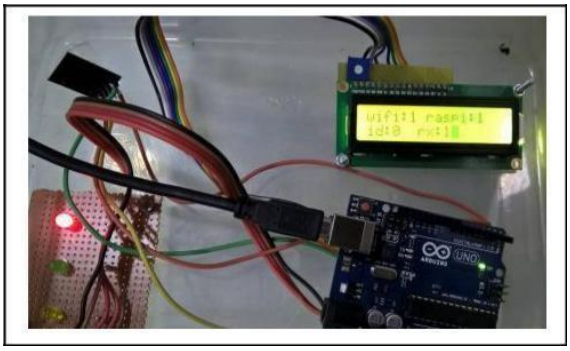


Fig.8. Results of data send through WI-FI

VI. Conclusion

This paper discusses the IOT based traffic control system in which signal timings are updated based on the vehicle density on the road. The proposed system is implemented using raspberry pi and Arduino. MATLAB Simulink support package leads to easy implementation of the system. Density measurement is performed by raspberry pi. Proposed system gives good performance and helps to reduce traffic congestion. Results shows vehicle counting in real time. The proposed system is suitable for real time vehicle management.

References

[1] Sk Riyazhussain, Riyazhussain, C.R.S. Lokesh, P.Vamsikrishna, Goli Rohan, "Raspberry Pi Controlled Traffic Density Monitoring System", *IEEE WiSPNET*, 2016.

[2] M. Vidhyia, S. Elayaraja, M.Anitha, 4M.Divya and S. Divya Barathi,"Traffic Light Control System Using Raspberry-PI", *Asian Journal of Electrical Sciences,* ISSN: 2249 - 6297 Vol. 5 No. 1, 2016, pp.8-12.

[3] Sony Francis, Prof. Sunitha Beevi K ,"Intelligent Traffic Control using Raspberry PI", *International Journal of Electronics, Electrical andComputational System, IJEECS*, ISSN 2348-117X, Volume 5, Issue 6, June 2016.

[4] K. Vidhya, A. Bazila Banu, "Density Based Traffic Signal System", *International Journal of Innovative Research in Science,Engineering and Technology*, Volume 3, Special Issue 3, March 2014

[5] Roxanne Hawi, George Okeyo, Michael Kimwele, "Techniques for Smart Traffic Control: An In-depth Review*", International Journal of Computer Applications Technology and Research,* Volume 4– Issue 7, 566 - 573, 2015, ISSN: 2319–8656.

[6] Aizuddin bin Onn, Safyzan Salim, Muhammad Shukri Ahmad, Muhammad Mahadi Abdul Jamil, "Development of a Network-Enabled Traffic Light System", *IEEE International Conference on Control System, Computing and Engineering*, Penang, Malaysia, 28 - 30 November 2014.

[7] Mr. P.Sivasankar., B.Brindhavathy "IOT Based Traffic Monitoring using Raspberry Pi", *International Journal of Research in Engineering and Technology.*",Vol.1,No.7, April 2016.

[8] S.Lokesh , T.Prahlad Reddy " An Adaptive Traffic Control System Using Raspberry PI", *International journal of engineering sciences& research technology*,June 2014.

[9] Mohmmad Rubaiyat Tanvir, Md.Asif Shahjalal, Nowroz Farhan Nur,"Design of IoT based Autonomous Vehicle with the aid of Computer Vision", *International Conference on Electrical, Computer and Communication Engineering (ECCE),* February 16-18, 2017,Cox's Bazar, Bangladesh,IEEE.

Authorized licensed use limited to: VIT University. Downloaded on August 15,2025 at 07:08:39 UTC from IEEE Xplore. Restrictions apply.